Research activity

Derivatives and reading graphs — 35 min.

Consider the function f defined on $\mathbb{R}$ by $f(x) = x^3 - 3x$. Denote its graph by $\mathcal{C}_f$.

1. Calculate $f'(x)$ and $f''(x)$, then draw up their sign tables.
2. Deduce the variations of f and its intervals of convexity and concavity.
3. Determine the coordinates of the point of inflection of $\mathcal{C}_f$ and an equation of the tangent at this point.
4. Using the limits, the extrema and the intermediate value theorem, determine the number of real solutions of $f(x) = 1$.
5. Explain, from this example, why continuity is not enough to ensure that a solution is unique.